

Virtualized Storage

COMING WITH Microsoft's Windows 8

<There's a lot to like, and a lot to hate; but there's one brilliant feature in Windows 8 that's hard to neglect, for now at least.> **By Kshitij Sobti**

There are several new features that get added to each Windows version that doesn't get the highlight it deserves. In Windows 7, it was the ability to mount and boot from a virtual hard drive. Windows 8 will feature a major upgrade to the way Windows can handle storage devices, with a new feature called Storage Spaces. This should end a majority of the storage woes that most people have. There are three key "features" of Windows Storage Spaces that we should all be excited about, and they are:

- Storage Pooling
- Thin Provisioning
- Resiliency to Hardware Failures

What's interesting isn't just the availability of these three features, but also how they work together. Perhaps you're already familiar with these terms and are already excited about what Windows 8 could hold. We won't hold you back any longer.

Storage Pooling

Hard drives have this way of shrinking over time, don't they? Perhaps not in actual capacity, but in the adjectives used to describe them. In just a few years' time, 80 GB hard drives went from "Wow! That's a lot!" to "Where do I backup the other half of this Blu-ray rip?"

One of the problems with the way storage works in Windows right now is that adding a new hard drive doesn't give you more space in each partition, instead it gives you another blank area to create new partitions in. This means reinstalling Windows, and/or moving all your documents and files over to a new hard drive; a tedious process. Over time, one tends to replace hard drives, even if they are working fine and is under warranty. It is just more convenient to have one larger hard drive than a few smaller ones having a few partitions each.

A better solution would be to pool, or combine all your hard drives into one storage space, and create partitions — without worrying about where one hard drive ends, and another begins. This is what storage pooling is.

In Windows 8, you'll be able to simply add a new hard drive, and have its capacity available to you in one combined pool that encompasses all your hard drives. This way you can have a 600 GB partition across two 500 GB hard drives. Windows 8, however, will enable something even better, as you find out next.

Thin Provisioning

Let's say you bought a 500 GB hard drive a few years ago, and created a 100 GB Windows partition, a 200 GB partition for your videos, a 100 GB partition for your music, and another 100 GB for your documents.

Now, your partition for videos could get exhausted, while your partition meant for documents could have a lot of free space. There isn't much you can do to expand your videos partition to take up the space from the documents partition. You could at most take some space from the Windows partition.

If you have pooled your storage in Windows 8, it will offer a new way to manage storage. You will be able to create partitions based on the amount of space required for them, instead of the amount of space available.

Thin provisioning, means to give the appearance of having more resources than are actually available. In Windows Storage Spaces this means you will be able to create a 20 TB videos partition, a 1 TB music partition, and a 500 GB documents partition, all on a 500 GB hard drive. All of this storage won't really be available of course! There is no new magic compression scheme in use here. The partition capacities are mostly meaningless here.

You can start with a 500 GB hard drive, with the partition structure described. As your usage reaches 500 GB, you can just add another hard drive, say of 2 TB. Now Windows will transparently start using space from both hard drives till you start reaching 2.5 TB usage. You can further extend your available capacity by adding more hard drives.

No need to worry about running out of space in one partition and having to re-partition. Windows will only use the storage capacity as it is available, and allocate it to the storage volume that needs it. Due to the inherent inflexibility of the older system, if you misestimate, and end up filling your documents drive before your videos drive, you have a problem. With thin provisioning, only the drives actually storing something will take up space so you might as well end up using all the 500 GB of your documents volume while your videos volume is empty. Extending the size of existing volumes is possible, so there's nothing to worry about, even if you use up a volume entirely.

Data Resiliency

Hard drive failures can be a tragic occurrence indeed, despite your best efforts. To protect from hardware failures, your data should be stored redundantly.

Storage Spaces will help here as well, by supporting different levels of resiliency. For this, Storage Spaces support data mirroring across multiple separate hard drives. Data on storage volumes that have mirroring enabled will automatically be stored on multiple hard drives. You can even select how much redundancy the data should have, either two separate hard drives, or three. At any point you could then pull out any one of the hard drives — or if one of your drives should fail — your data would continue to be accessible. The faulty hard drive can be replaced with a new one, and windows will